1. What is Prestige ONE & How It Works

Prestige ONE is a mobile-first, enterprise-grade Progressive Web Application (PWA) purpose-built for premium international academies. Developed for FLECTERE, it unifies student admissions, classroom registers, academic reporting, and tuition billing into a single authoritative digital platform.

🏛️ The Fictional Demonstration Academy

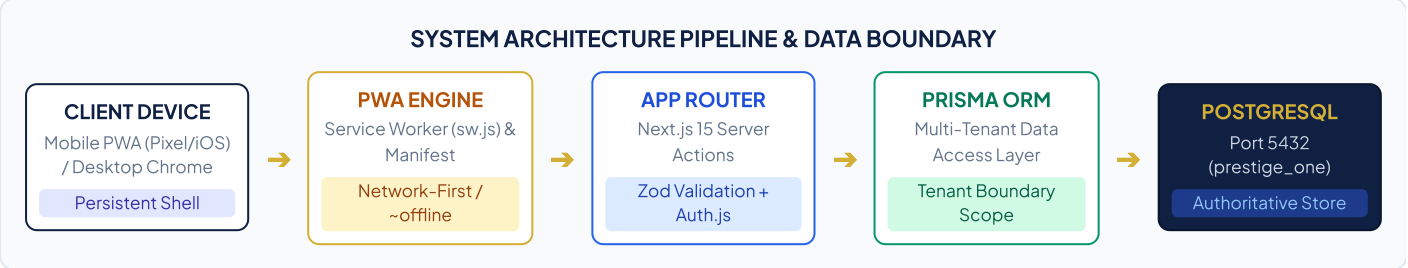
Prestige International Academy
- Location: Sommerschield Executive Quarter, Maputo
- Academic Year: **2026 (Term 1 Active)**
- Base Currency: **Mozambican Metical (MZN)**
- Live Class Cohorts: **Grade 7A, 7B, and 8A**
- Mission: Providing seamless digital continuity across leadership, teachers, parents, and students.

⚡ Core Technological Pillars

- **Zero Mock Data**: Native PostgreSQL database with strict relational constraints and audit logs.
- **Persistent App Shell**: Header, sidebar, and mobile nav survive internal client-side navigation without unmounting.
- **Zero-Data-Loss PWA**: Custom service worker precaches offline fallbacks while ensuring authenticated data is never unsafely cached.
- **Dual-Screen Fluidity**: Fully responsive across 375px mobile phones, tablets, and wide executive dashboards.

2. Platform Architecture & Data Flow

The platform guarantees that an action taken on any device (e.g., enrolling a student or recording an attendance roll) writes durably to PostgreSQL and synchronizes across all active directories and class rosters instantly.



3. Key Modules Available for Demonstration

📊 Executive Command Centre

URL: `/dashboard`

Live enrollment counts, term attendance metrics, fee collections in MZN, and recent institution-wide activity audit feeds.

🎓 Student Directory & Master Profile

URL: `/students` & `/students/[id]`

Searchable directory, real-time status badges, emergency contacts, medical records, and inline record mutation.

📋 Class Rosters & Daily Register

URL: `/classes` & `/attendance`

Classroom sections (Grade 7A, 7B, 8A) linked directly to student enrollment, plus 1-click morning attendance register.

P1 Demonstration Credentials & Personas

Prestige International Academy • Access Control Matrix

Section 2: Who's Who & What's What
Portal URL:
`http://localhost:3000/login`

2. Who's Who & What's What — Demo Personas

Prestige ONE incorporates strict role-based access control (RBAC). For testing convenience, the login page features **1-Click Persona Quick-Fill Buttons** that instantly populate the form with verified demo credentials.

Persona & Role	Login Email	Password	Key Permissions & Scope	Primary Screens
Catarina Nhaca INSTITUTIONAL ADMIN	`admin@prestige.co.mz`	`PrestigeAdmin2026!`	Full write access: Student enrollment, editing profiles, managing classes & financial records.	`/dashboard` `/students` `/students/new`
Dr. Armando Guambe ACADEMIC DIRECTOR	`director@prestige.co.mz`	`PrestigeDirector2026!`	Executive oversight: Aggregated KPIs, institutional attendance trends, faculty performance.	`/dashboard` `/classes` `/finance`
Helena Cossa SENIOR TEACHER	`teacher@prestige.co.mz`	`PrestigeTeacher2026!`	Classroom execution: Section Grade 7A roster, daily morning roll call, marking attendance.	`/attendance` `/classes/c-7a` `/students`
Teresa Sitoe FAMILY GUARDIAN	`guardian@prestige.co.mz`	`PrestigeGuardian2026!`	Parent portal: View linked child (Bernardo Sitoe), fee statements, attendance confirmation.	`/students/[id]` `/finance`
Bernardo Sitoe ENROLLED STUDENT	`student@prestige.co.mz`	`PrestigeStudent2026!`	Student portal: View enrolled class section, daily timetable, and personal academic profile.	`/dashboard` `/students/[id]`

Role Capabilities Breakdown ("What's What")

🏛️ Administrator Workflow (The Primary Slice)

- Navigate to **Student Directory** (`/students`) to see all 6 seeded students.
- Click **"Enroll New Student"** button to open the responsive multi-section form.
- Enter student demographics, select class **Grade 7A**, input guardian details.
- Submit: notice the instantaneous button lock (preventing double submissions), instant redirect, and persistent toast notification.

👩‍🏫 Teacher Workflow (Attendance Register)

- Navigate to **Daily Register** (`/attendance`).
- Select section (**Grade 7A**) and toggle any student between **Present, Late, Absent, Excused**.
- Status updates save asynchronously to PostgreSQL without page reloads.
- Navigate to **Classes** (`/classes`) to verify student counts update immediately.

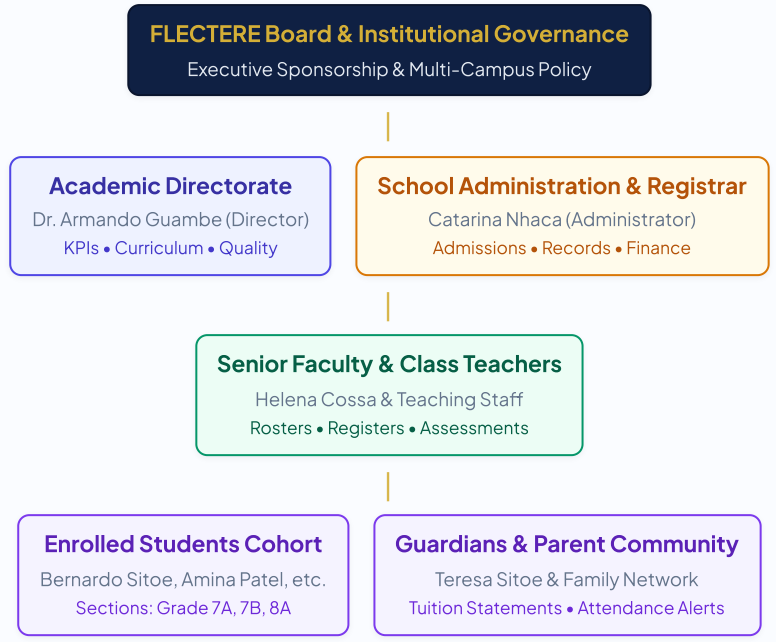
💡 **Client Demo Tip:** Open an incognito browser window or smartphone beside your desktop screen! Log into the mobile view as *Helena Cossa (Teacher)* and desktop view as *Catarina Nhaca (Admin)* to showcase real-time cross-device

data consistency backed by PostgreSQL.

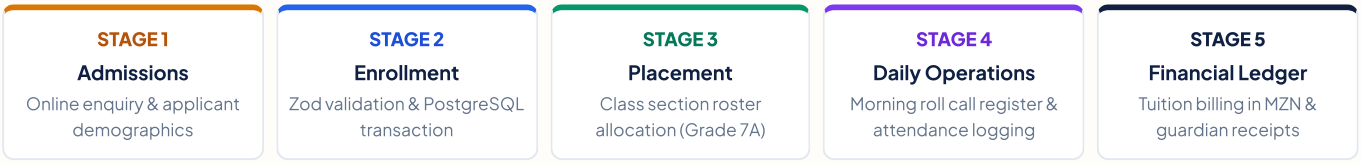
3. Visualizing How Prestige ONE Operates

The diagrams below illustrate the institutional hierarchy, the student operational lifecycle, and the role boundaries enforced within Prestige ONE.

A. Institutional Governance & User Organogram



B. Student Lifecycle & Operational Data Pipeline



C. Progressive Web App (PWA) Offline Reserve Mechanism

🛡️ **Client Continuity & Integrity Guarantee:**

Prestige ONE incorporates an intelligent Service Worker (`sw.js`). If internet connectivity drops during school operations:

1. The persistent header immediately updates the network pill from • Online to • Offline Mode.
2. Cached static assets and previously loaded screens remain responsive without blank white screens.
3. New navigations gracefully serve the **Prestige ONE Offline Reserve Screen** (`/~offline`) informing the user that data edits require an authoritative PostgreSQL connection.
4. When connection restores, the status pill auto-recovers to Online without losing local page state.

P1 Live Demonstration Script (Step-by-Step)

The 12-Step Primary Vertical Slice & Testing Verification

Section 4: Demo Script
Verified by 40 Playwright Scenarios

4. Step-by-Step Demonstration Walkthrough

Follow this exact 12-step sequence to deliver a flawless client demonstration of the primary workflow: **Student Enrollment and Institutional Record Synchronization**.

Step 1 **Sign In as Administrator**: Visit `http://localhost:3000/login`. Click the *"Admin"* button to fill `admin@prestige.co.mz` / `PrestigeAdmin2026!`. Click **Sign In**.

Step 2 **Inspect Executive Dashboard**: Arrive at `/dashboard`. Point out the live metrics: *Active Students (6)*, *Term Attendance (94.2%)*, *Tuition Collected (82.5% MZN)*.

Step 3 **Open Student Directory**: Click *Students* in the sidebar or bottom mobile nav (`/students`). Notice instant client-side transition without page reload. Filter or search existing students.

Step 4 **Initiate Enrollment**: Click **"+ Enroll New Student"** button to navigate to `/students/new`.

Step 5 **Input Student Details**: Enter Admission No (e.g., `PIA-2026-099`), First Name (*Zayd*), Last Name (*Mabunda*), Date of Birth, Gender, Select Class Section: **Grade 7A**.

Step 6 **Input Guardian & Medical Profile**: Enter Guardian Name (*Salimo Mabunda*), Email (*salimo.m@example.com*), Phone (*+258 84 900 1122*), and Medical Notes (*No known allergies*).

Step 7 **Submit Form with Concurrency Lock**: Click **"Complete Student Enrollment"**. Observe the button immediately disable to prevent duplicate submissions, followed by a green toast confirmation.

Step 8 **Verify Master Profile**: You are redirected to `/students/[id]`. Highlight the generated student banner, emergency phone, residential address, and active class enrollment.

Step 9 **Cross-Section Verification**: Click *Classes* in navigation, then open **Grade 7A** (`/classes/[id]`). The new student appears directly in the section roster with full details!

Step 10 **Edit Student Record**: Return to the student profile, click **"Edit Record"** (`/students/[id]/edit`). Modify Emergency Phone to `+258 84 999 8877` and click **Save Changes**.

Step 11 **Verify Synchronization**: Confirm the updated phone number renders immediately on both the student profile and the class roster.

Step 12 **Prove Durable Persistence**: Perform a hard browser reload (`Ctrl + R`) or log out and log back in. The record is permanently preserved in PostgreSQL!

5. Quick Testing & Verification Commands

Start Dev Server:	`npm run dev` (Available at `http://localhost:3000`)
Run Unit Tests:	`npm run test` (8 Vitest unit tests pass in ~5s)
Run E2E Browser Suite:	`npx playwright test` (40 tests across Desktop & Mobile Chrome pass)
Production Build:	`npm run build` (Strict Next.js App Router production compiler)